CODERED '26 | INDUSTRY CHALLENGE BRIEF

Problem Statement 2
The Perfect Fit

Instant ATS scoring, one-page resume optimisation and auto-generated portfolios

OBJECTIVE Address the challenges job applicants face with single-page resume formatting and ATS compatibility through an automated platform that provides instant ATS scoring, actionable improvement parameters and an automatically generated portfolio website.

1. Challenge Overview

Company / Partner Name: MyStartupWave

Challenge Title: The Perfect Fit

Industry / Track: AI for Careers / HR Technology

Challenge Contact: MyStartupWave Challenge Desk (mystartupwave@gmail.com)

Challenge Type: New Product Development

Brief Summary: Job seekers routinely lose opportunities because their resumes are rejected by Applicant Tracking Systems (ATS) or fail to fit a clean single-page format. Teams will build an automated platform that parses any resume, scores it for ATS compatibility against a specific job description, gives precise and actionable improvement guidance, reformats it into a polished one-page layout, and automatically generates a personal portfolio website.

2. Industry Context

Most mid-size and large employers, and a growing share of startups, screen applications through Applicant Tracking Systems before a human ever reads them. Campus placements and off-campus hiring in India attract very large numbers of applicants per role, and recruiters typically spend only seconds on an initial resume scan.

Many qualified candidates, especially students and early-career professionals, are filtered out because of parsing failures (tables, columns, graphics, unusual fonts), missing keywords or poor structure rather than a lack of skill. Recruiters also increasingly expect a portfolio or online presence. Advances in document parsing and language models now make it possible to give every candidate recruiter-grade feedback instantly.

3. Problem Statement

- **Problem and who is affected:** Students, freshers and career switchers apply to many roles without knowing why their resume is rejected. Recruiters also lose strong candidates whose resumes are parsed poorly.
- **When and where it occurs:** At every application, especially in high-volume campus placement drives and job-portal applications, where the resume must be tailored to each job description under time pressure.
- **Limitations of current solutions:** Existing ATS checkers give a vague score with generic tips, often hide detailed feedback behind paywalls, ignore the specific job description, or do not explain how the score is calculated. Resume builders produce templates that are not always ATS-safe and rarely enforce a single page. Portfolio sites must be built separately and by hand.

- **Why it is worth solving:** Candidates often submit dozens of applications per offer; placement cells and recruiters repeatedly cite formatting and keyword gaps as common rejection reasons; and the market for paid resume review and builder tools is large and growing, signalling strong willingness to pay for better outcomes.

4. Target Users and Stakeholders

Primary users: Students, freshers and early-career professionals applying through ATS-driven hiring processes.

Secondary users / stakeholders: University placement cells, career counsellors, recruiters and hiring managers, bootcamps and training institutes.

Current user journey or workflow: Download a template → fill it in Word or Canva → guess keywords from the job description → upload to a free ATS checker → receive a generic score → edit manually → struggle to fit everything on one page → build a portfolio site separately, or skip it.

Key user pain points: Opaque scoring; generic advice; formatting that breaks parsing; content spilling beyond one page; repetitive tailoring for every job; and no time or skill to build a portfolio website.

5. Product Opportunity

- **What users should accomplish:** Upload a resume (PDF or DOCX) and a target job description and receive, within seconds, an explainable ATS score with a parameter-by-parameter breakdown.
- **Meaningful improvement:** Specific, line-level improvements (rewritten bullets, missing keywords, quantified impact, section order) and an exported ATS-safe, single-page resume that keeps all critical content.
- **What makes it different:** Transparent, deterministic scoring tied to a real job description, plus a professional, mobile-responsive portfolio generated from the same data in one click, keeping resume and portfolio in sync.

6. Expected Product Capabilities

Must-have capabilities:

- Robust parsing of PDF and DOCX resumes, including multi-column layouts, tables and non-standard section headings, into a structured schema (contact, education, experience, projects, skills, certifications).
- A job-description-aware ATS score (0–100) with a transparent breakdown of at least eight parameters, e.g., keyword match, skills coverage, section completeness, formatting/parseability, action verbs, quantified achievements, length/density and readability.
- Actionable, line-level suggestions rather than generic tips, including AI-rewritten bullet points that the user can accept, edit or reject.
- Automatic single-page optimisation: intelligent condensing, prioritisation and layout so the final resume fits exactly one page while remaining ATS-safe.
- Export to ATS-friendly PDF and DOCX.
- One-click generation of a responsive portfolio website from the resume data, published at a shareable URL.

Nice-to-have capabilities:

- "Before vs. after" score comparison and version history for each job application.
- Multiple portfolio themes, custom-domain support and import from GitHub or LinkedIn.
- Fairness checks that flag details which should not appear on a resume (e.g., photo, age, marital status).
- Support for Indian and international resume conventions and regional-language interfaces.
- A recruiter view to bulk-score resumes against a job description.

Expected user interactions: Upload resume → paste or link a job description → view the score dashboard and parameter breakdown → accept or edit suggestions inline and watch the score update live → download the optimised one-page resume → publish the portfolio and share its link.

Expected demonstration / deliverable: A live end-to-end flow using at least three resumes of different quality and format (including a two-column, graphics-heavy resume), each scored against a real job description, showing the score improvement, a one-page export and a live portfolio URL.

7. Technical and Operational Constraints

Technical or platform constraints: Must work in a modern browser without installation. Must handle text-based PDF and DOCX; scanned or image-only PDFs must either be supported through OCR or rejected with a clear message. Generated resumes must use ATS-safe layouts (single column, standard fonts, no text inside images, headers or footers).

Performance, accuracy, or latency requirements: Parsing and scoring in under 10 seconds per resume. Field-level parsing accuracy of at least 90% on the judges' test resumes. Scores must be deterministic: the same resume and job description must return the same score (±1). Suggestions must never invent experience, skills, employers or metrics the user did not provide.

Security, privacy, or compliance requirements: Resumes contain personal data. Encrypt data in transit (HTTPS), let users delete their data, do not use uploaded resumes for model training without explicit consent, and align with the principles of India's Digital Personal Data Protection Act, 2023. Portfolio publishing must be opt-in, with user control over which contact details are public.

Hardware, connectivity, or deployment limitations: Must work on low-bandwidth mobile connections. Portfolio sites should be lightweight static pages.

Budget or resource constraints: Should operate within free tiers or low-cost inference. Teams must document the processing cost per resume.

Domain-specific safety requirements: No fabrication or exaggeration of credentials; AI-generated rewrites must be clearly labelled; no recommendations that encourage keyword stuffing or hidden text.

8. Available Resources and Support

Resources marked "Confirmed" are guaranteed by the partner. "Permitted" items may be used by teams at their own discretion and cost. "Under consideration" items are not yet confirmed.

APIs, SDKs, developer tools, or technology credits: Confirmed: none provided by the partner. Permitted: open-source parsers and language models, or generative AI APIs available to the team.

Sample datasets or permitted data sources: Team members' own resumes (with consent), synthetic resumes and publicly available job descriptions. Public resume datasets may be used where their licences permit. Judges will evaluate using a hidden test set of resumes and job descriptions.

Sandbox environments or test infrastructure: Not provided. Teams use their own hosting (e.g., Vercel, Netlify, GitHub Pages for portfolios).

Documentation and reference materials: Publicly available ATS vendor documentation and recruiter guidance on resume parsing; Digital Personal Data Protection Act, 2023.

Mentors, domain experts, or technical sessions: Confirmed: MyStartupWave engineers during the scheduled mentoring rounds (subject to the event schedule).

Other resources or support: Under consideration: a feedback session with campus placement coordinators.

9. Business and Adoption Context

Intended customer or beneficiary: Students and job seekers (B2C); universities, placement cells and training institutes (B2B / B2B2C).

Current alternatives or competing solutions: Online ATS checkers (e.g., Jobscan, Resume Worded), resume builders (e.g., Canva, Zety, Novorésumé) and general-purpose website builders used for portfolios (e.g., Wix, Notion).

Potential value created: Higher shortlisting rates for candidates, time saved on tailoring every application, fewer qualified candidates lost to parsing errors, and a professional online presence with no extra effort.

Potential adoption pathway: Freemium for individual students; institutional licences for colleges and placement cells; partnerships with hiring platforms.

Commercial or operational considerations: Language-model inference cost per analysis, portfolio hosting cost, data-protection obligations and integration with university placement portals.

Known assumptions or uncertainties: ATS behaviour varies by vendor and is not fully public, so the score approximates common ATS logic; score improvements are assumed to correlate with better shortlisting outcomes.

10. Success Indicators

Functional outcomes: Correct parsing, scoring, suggestions, one-page export and portfolio generation for all judge test resumes, including complex layouts.

User or customer outcomes: Average ATS score increase of at least 25 points after accepting suggestions; users can go from upload to a published portfolio in under 5 minutes.

Technical performance indicators: ≥ 90% field-level parsing accuracy; < 10 seconds per analysis; 100% of exported resumes fit exactly one page and re-parse cleanly through the team’s own parser and at least one external parser.

Business or operational indicators: Clear unit economics (cost per analysis) and an adoption plan for at least one institutional customer segment.

Safety, reliability, or compliance indicators: Zero fabricated content in suggestions during judge spot-checks; a working data-deletion flow.

11. Scope and Boundaries

In scope: Resume parsing, job-description-aware ATS scoring, explainable feedback, AI rewrite suggestions, single-page optimisation, export, and portfolio generation and hosting.

Out of scope: Integration with real employers’ ATS systems, applying to jobs on users’ behalf, guaranteeing interview outcomes, and cover-letter generation (optional only).

Permitted assumptions or simplifications: Scoring may be based on a documented model of common ATS behaviour rather than any single vendor’s proprietary algorithm. English-language resumes are sufficient for the core demonstration.

Dependencies (systems, approvals, resources): A document-parsing approach, a language model for rewriting, and static hosting for portfolios.

12. Industry Engagement

Will the company provide mentors or subject-matter experts? Yes. Engineers from MyStartupWave will be available during the scheduled mentoring rounds for architecture, product and feasibility reviews.

Will teams receive feedback during development? Yes. Teams will receive structured feedback at the designated mid-event checkpoint(s), focused on scope, technical approach and progress against the success indicators.

Will the company participate in evaluation or demonstrations? Yes. MyStartupWave representatives will attend the final demonstrations and take part in evaluation alongside the CODERED ’26 jury.

Is further validation, pilot discussion, or product exploration possible after the event? Possible. Teams with exceptional solutions may be invited to a follow-up discussion on further validation, mentoring or pilot exploration.

Any follow-on engagement is subject to mutual interest and separate agreement.

13. Intellectual Property and Participation Terms

The following terms apply to all participating teams and are communicated before development begins. Participation does not imply any automatic transfer of ownership or commercial rights; any subsequent arrangement must be agreed separately.

Data and resource usage conditions: Use only synthetic resumes, public job descriptions, openly licensed datasets, or resumes shared with the explicit consent of their owners. Do not scrape personal data from job portals or social networks.

Confidentiality requirements, if any: None. This brief and any materials shared for this challenge are non-confidential.

Intellectual property and ownership terms: Teams retain full ownership of the code, designs, models and other intellectual property they create. Participation does not transfer any ownership or commercial rights to MyStartupWave.

Restrictions on using company names, data, or materials: Do not use MyStartupWave's name or branding without written permission. Names of third-party ATS vendors or job portals may be referenced only descriptively.

Terms governing subsequent pilots, licensing, investment, employment, or commercial collaboration: Any pilot, licensing, investment, employment or commercial collaboration after the event is subject to mutual interest and a separate written agreement.

14. Partner Contact and Clarifications

Primary contact: MyStartupWave Challenge Desk

Email: mystartupwave@gmail.com

Reference documentation / resource links: This challenge brief; Digital Personal Data Protection Act, 2023; public ATS vendor documentation on resume parsing.

Clarification process: Send questions in writing to the challenge contact or raise them through the official CODERED '26 helpdesk. Answers that affect scope or evaluation will be shared with all teams on this challenge at the same time, to keep the competition fair.

Submission Checklist

- ☑ The problem is real, relevant, and clearly stated.
- ☑ Target users and desired outcomes are identified.
- ☑ Teams have freedom to develop their own solutions.
- ☑ A working prototype is reasonably achievable within the challenge timeline.
- ☑ Business value and adoption can be evaluated.
- ☑ Technical constraints and available resources are stated.
- ☑ Data, safety, confidentiality, and IP terms are clear.
- ☑ Evaluation expectations are communicated consistently.

THE GUIDING PRINCIPLE

The company defines the opportunity. The team defines the solution.